

Drug Discovery: Protein-Protein Interaction Prediction using Decoder-only Transformer Model

Mayank Maurya

School of Computing, DIT University-
Dehradun
Dehradun, India
email: 1000020804@dit.edu.in

Amit Chatterjee

School of Computing, DIT University-
Dehradun
Dehradun, India
email: amitchat.instru1992@gmail.com

Abstract—Understanding protein-protein interactions (PPIs) are essential for comprehending various biological processes, including disease causing microbes pathological mechanisms. With the rapid advancement in deep learning (DL) techniques, transformer-based models have shown promising results in analysing sequential data in natural language processing which may be extended to protein amino acid sequence. In this paper, a novel technique for efficiently and accurately predicting PPIs utilizing transformer models is proposed. In order to predict the capability of interaction, our proposed model quickly evaluates a pair of provided FASTA protein sequences as input leveraging decoder-only transformer model. This enables our model to predict PPIs with previously unprecedented accuracy. The findings reveal how this technique may be used for large-scale prediction tasks, allowing researchers to quickly and precisely discover possible new protein-drugs and targets. This finding has important ramifications as it may help find novel therapeutic protein-based medications by identifying proteins that interact with target proteins associated with diseases.

Keywords— *protein-protein interaction, decoder-only transformer model, FASTA, deep learning, amino acid sequence*

I. INTRODUCTION

Understanding PPIs is crucial for comprehending biological processes behind diseases, such as signal transmission, metabolism, and cell signalling pathways [1-5]. Finding possible therapeutic drugs and their targets, creating novel medications, and enhancing current therapies are all made possible by accurate PPI predictions. The intricacy and specificity of PPIs make their prediction from sequencing data a challenging task. Computational models have become an important tool for PPI prediction due to the rapid development of artificial intelligence (AI) and DL methods.

In this direction, the TUNA model, developed in 2024 [6], uses a Transformer architecture to predict PPIs from sequence data. It used ESM-2 embeddings and SNGP encoders on cross-species datasets like Bernett, achieving a balanced accuracy measure of 0.3 and achieving AUROC, AUPR, Balanced Accuracy, Precision, Recall, Specificity, and F1 score of 0.65. However, it requires a large amount of memory for embedding, making it computationally demanding. Paper [7] proposed DeepAraPPI a method for predicting PPIs in *Arabidopsis thaliana* using word2vec and domain2vec encoding-based models. It has shown impressive results on datasets like IntAct, DIP, MINT, and BioGRID. However, it struggles with unknown proteins and domain annotation quality. Despite its limitations, DeepAraPPI outperforms other

models with an overall score of 0.828. A 2023 study proposed ISPRE-SEQ, a deep neural network model that predicted interaction sites in protein sequences with high accuracy [8]. The model used ProtT5 and ESM1-b, an encoder-only transformer model trained on over 27 million sequences. On a dataset of 6,066+448 non-redundant protein chains, it achieved AUC of 0.82. However, the model relied on the quantity and quality of interaction data and sequence length. A pipeline that used structural similarity, sequence similarity, functional association data, and machine learning to predict PPIs across whole proteomes was suggested in [9]. PEPPI used threading-based interlay for quicker structural analysis, achieving high accuracy on a dataset of 638,600 connections. However, PEPPI's capacity to discover novel interactions was limited by its reliance on structural similarity-based predictions and not considering biological context, such as subcellular localization. Researchers have proposed various models for predicting PPI sites, with one recent study introducing HN-PPISP, a hybrid neural network that used Bi-LSTM for global features and MLP-Mixer and TextCNN for local features [10]. The method achieved high accuracy and precision in two datasets, but has two significant drawbacks: its complex model architecture and high processing resource requirements. However, HN-PPISP's effectiveness in extracting extended relationships suggests significant advancements over current models.

Although, previous models have shown great promise in predicting PPIs with high accuracy using sequence data, but they often rely on network information or 3D structure of proteins, which can hinder deeper understanding of proteins and PPIs. A lot of conventional machine learning (ML) techniques rely on 3D-structure data or protein network heuristics, which could result in biased models that don't completely comprehend the underlying features. These models often require extensive feature engineering and hyperparameter tuning, making them computationally expensive and time-consuming to train and evaluate. The existing transformer models proposed by in few existing research papers repurpose the Large Language Models (LLMs) designed for Natural Language Processing (NLP) which are computationally expensive and not specialised for this purpose. Furthermore, the complexity of PPI, including subcellular localization or temporal dynamics, could be missed by current methods.

Hence, to circumvent the drawbacks of existing techniques, the proposed study aims to apply a novel transformer model to PPI by considering them as a string of

amino acid sequences. This paper suggests a decoder-based transformer model, which contributes in following ways to the objective:

- The suggested model does not require other data sources such as protein networks or 3D structures because it is designed to predict PPIs using only two FASTA-based protein sequences as inputs.
- Model uses significantly less trainable weights which makes it faster than both conventional ML techniques and the most advanced DL models currently in literature.
- With a minimal amount of processing power, the model can manage large-scale prediction tasks.

To explore the use of transformer models in PPI prediction, understanding of few key topics are needed to be discussed:

A. Protein-Protein Interactions (PPI)

PPIs play a vital role in various biological functions, impacting signal transduction, metabolism, and cell signalling pathways. Understanding PPIs is crucial for understanding diseases like cancer, neurological disorders, and infectious diseases [2]. Factors like protein structure, post-translational changes, and environmental conditions influence these interactions. PPI research has significant implications for drug discovery.

B. FASTA-Protein Sequence

A common format in bioinformatics for representing protein sequences is FASTA (FAST-All) [3]. It is a crucial tool for many applications, including sequence alignment, multiple sequence comparison, and phylogenetic analysis, since it offers a clear and succinct method of representing amino acid sequences.

C. Transformer Models

Transformers, a deep neural network family, has advanced NLP applications like sentiment analysis and machine translation. Its ability to capture long-range connections between tokens, like amino acids, has made it popular in bioinformatics. Transformers have shown promising performance in grouping, regression, and sequence classification, making their application in PPI prediction promising [4].

D. Decoder-only Transformer Models for PPI

These neural network architectures, have shown superior performance in learning intricate patterns between long sequences e.g. protein sequences, without the need for both encoder-decoder structures [5], making them intriguing for PPI prediction. They can be trained with a range of datasets and objectives, making them a desirable for predicting PPIs.

II. METHODOLOGY

The proposed methodology for PPI prediction includes dataset description, preprocessing methods, architectural overview, and model construction, with visual aids like program flowcharts and model architecture diagrams ensuring transparency and reproducibility throughout the study.

A. Dataset description

The String-DB dataset is a prominent database of PPIs with experimental validation. The dataset's purpose, source,

size, data kinds, and usage issues are all summarized in this section.

1) Dataset Overview

59.3 million proteins and almost >20 billion interactions between them make up the String-DB dataset. Each protein is represented by an amino acid sequence in FASTA format, which serves as input for machine learning models. It was developed by combining pre-existing databases to offer a thorough resource for PPI research. Numerous research has made extensive use of this dataset to create machine learning models for PPI prediction.

Purpose: The String-DB dataset's objective is to offer a trustworthy source of empirically verified PPIs for AI model testing and training. Researchers want to create more accurate prediction models and get a deeper understanding of the underlying mechanics of PPIs by examining this dataset.

Source: STRING v12: A comprehensive database of known and predicted PPIs.

2) Data Types

The dataset comprises an abundance of PPI-related data categories, including:

- Protein names and IDs: Each protein is given a human-readable name and ID; data type: string, e.g. ENSP00000380432 (Insulin).
- Amino acid sequences: Unique sequences that identify specific proteins, data type: string, e.g.:
MALWMRLLPLLALLALWGPDPAAAFVNQHLCGSHLVEALYLVCGERGFFYTPKTRREAEDLQVQGVELGGGPGAGSLQPLALEGSLQKRGIV EQCCTSICSLYQLENYCN*.
- Amino acid sequences chunk: The amino acid sequence is divided into chunks for model to comprehend it. Data type: array of string, e.g. ['M', 'A', 'L', 'W', 'M', 'R', 'L', 'L', 'P', 'L', 'L', 'A', 'L', 'L', 'A', 'L', 'W', 'G', 'P', 'D', 'P', 'A', 'A', 'F', 'V', 'N', 'Q', 'H', 'L', 'C', 'G', 'S', 'H', 'L', 'V', 'E', 'A', 'L', 'Y', 'L', 'V', 'C', 'G', 'E', 'R', 'G', 'F', 'F', 'Y', 'T', 'P', 'K', 'T', 'R', 'E', 'A', 'E', 'D', 'L', 'Q', 'V', 'Q', 'V', 'E', 'L', 'G', 'G', 'G', 'P', 'G', 'A', 'G', 'S', 'L', 'Q', 'P', 'L', 'A', 'L', 'E', 'G', 'S', 'L', 'Q', 'K', 'R', 'G', 'I', 'V', 'E', 'Q', 'C', 'T', 'S', 'I', 'C', 'S', 'L', 'Y', 'Q', 'L', 'E', 'N', 'Y', 'C', 'N', '*']
- Confidence scores: Measures of confidence in the accuracy of each PPI. data type: float.

3) Usage

This database is used by researchers for numerous kinds of objectives, including:

- Predicting PPIs using AI-ML-DL algorithms.
- Development of novel techniques for identifying protein functional modules.
- Investigating the connection between the sequential structure and functions of proteins.

4) Challenges

Using this dataset presented several difficulties, such as the following:

- No pre-packaged dataset was available, and the data of all required species had to be aggregated iteratively.
- It was necessary to download the protein FASTA sequencing dataset separately.

- There isn't an existing library that can automatically establish a connection between the FASTA representation of a protein pair and its score.

B. Processing

1) Database Acquisition Section

a) STEP 1: Database Selection

Selecting an appropriate database for PPI prediction is the first stage in the database acquisition section. As shown in the Figure 1 flowchart, the source database is STRING v12, which offers a comprehensive set of known and predicted PPIs, is employed. The significance of choosing a reliable and accurate dataset for model training is very important. In this case, the extensive collection of PPIs in STRING v12 provides a great starting point for creating predictive algorithms.

b) STEP 2: Species Selection

This step involves the selection of the species for the research as we may not do the research on all available species in the biosphere because of resource limitation and time constraints. We need to select the species carefully in alignment to our research objective. For this research, top 10 species (subjective) from perspective of drug discovery were selected as shown in following Table I.

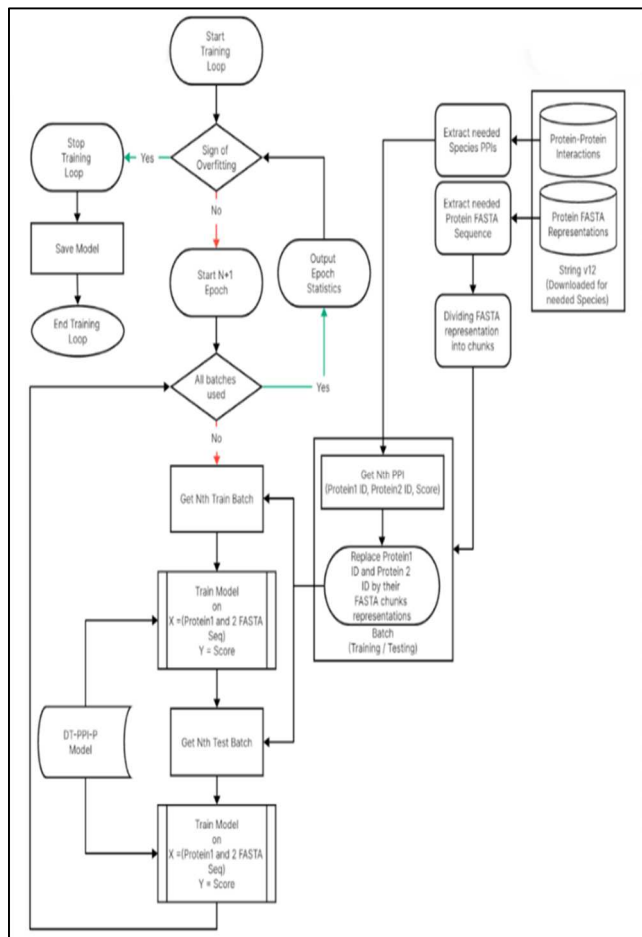


Fig. 1. Flowchart of program overview.

Table I. Selected 10 species for the research

#tax on id	Official name NCBI	Common Name	domain	Reason
9606	Homo Sapiens	Human	Eukaryotes	The most relevant for human health, drug development, and clinical applications.

10090	Mus Musculus	Mouse	Eukaryotes	Mice are widely used as model organisms in biomedical research.
10116	Rattus Norvegicus	Rat	Eukaryotes	Commonly used in pharmacology and toxicology studies for preclinical drug testing.
7955	Danio Rerio	Zebrafish	Eukaryotes	Frequently used in drug discovery for genetic and developmental studies.
6239	Caenorhabditis elegans	C. elegans	Eukaryotes	A simple model organism with a well-mapped genome and PPI network, useful for understanding fundamental biological processes relevant to drug targets.
7227	Drosophila Melanogaster	Fruit fly	Eukaryotes	Extensively used in the study of genetics and molecular biology, including human disease genes.
4932	Saccharomyces cerevisiae	Yeast	Eukaryotes	Well-studied PPI network often used in high-throughput screens for drug targets, particularly in cancer and metabolic disorders.
155864	Escherichia coli	Bacteria	Bacteria	Widely used in biotechnology and pharmaceuticals; E. coli PPIs help understand bacterial processes important for antibiotic drug development.
36329	Plasmodium falciparum	Malaria parasite	Eukaryotes	Critical target for antimalarial drug discovery, a major global health concern.
83332	Mycobacterium tuberculosis	Tuberculosis bacteria	Bacteria	Major global health concern; studying PPI networks helps in the discovery of novel antibiotics for treatment.

c) STEP 3: PPI Information gathering

The selection process involved obtaining PPIs data from the String database v12 for each species. The protein pairs with a combined confidence score of >0.4 selected as interactions and the rest as no interactions, as inspired from [11]. To balance the dataset and avoid biases, pruning was done and equal numbers of 'interaction' and 'no interaction' data-points were kept.

d) STEP 4: Protein FASTA Sequence Extraction

The protein pairs are presented as ENSEMBL IDs which are not compatible with the model's input requirements. To address this, unique amino acid sequences corresponding to individual proteins were extracted, aggregated, and mapped from String DB v12 in FASTA format, facilitating their translation into the predictive framework.

e) STEP 5: Representation Generation

Once necessary protein sequences are obtained, they need to be represented in a format that can be used for prediction. The FASTA representations are broken into chunks of amino acids:

Monopeptide: A single amino acid.

Dipeptide: Two amino acids linked by a peptide bond.

Tripeptide: Three amino acids linked by two or three peptide bonds.

Emphasizing the significance of converting sequences into usable formats. The conversion process enables models to learn from the provided data and generate accurate predictions more efficiently.

f) STEP 6: Training Loop

The training loop begins with variable initialization and loading of required datasets batches following data collection and representation development.

g) STEP 7: Training Tests

To ensure models learns correctly, a number of tests are made on various metrics during this stage, including overfitting, mode collapse, and gradient explosion detection.

h) STEP 8: Stopping Training Loop

When no further improvements are seen, the training loop stops marking training completion milestone, marking an important stage in developing predictive models.

i) STEP 9: Model Saving

The final prediction model is saved for future use and deployment as a '.keras' file after successfully completing all necessary steps without errors.

j) STEP 10: Output

The saved model is loaded and tested against an unseen test data and the final evaluation is gained by evaluating the model prediction on metrics, accuracy of 84.7% is obtained. Our predictive analysis tool is now prepared for further use by displaying final prediction results.

C. Model Architecture

The DT-PPI-PM is a decoder-only transformer model designed for protein-protein interaction prediction using sequence data. It embeds two input sequences (Protein_1 and Protein_2) into high-dimensional spaces and adds positional encoding. The model employs eight stacked decoder layers that learn intricate linkages and dependencies within and between protein sequences. The decoder layers iteratively improve comprehension of protein sequences. The final dense output layer generates a binary classification output by predicting protein interaction possibilities. This model leverages cross attention mechanisms for high accuracy.

1) Input Layers

These are the place holder layers for provided protein pair(s).

- Protein_1: A single input layer for the first protein sequence, shape of (None, MX_SEQ_LEN), representing the length of sequences.
- Protein_2: Another single input layer for the second protein sequence with the same characteristics as above.

MX_SEQ_LEN, maximum sequence length is 174, but the maximum amino acid sequence length is 512 as we divided amino acids into chunks of 1, 2, 3 as per need to fit it in 174 maximum sequence length. Ideally it could've been 522 (174 * 3) but extra symbols, SOS (Start of Sequence), * (stop codon), and EOS (End of Sequence), consumed the remaining space.

2) Embedding Layers

Two embedding layers are used to convert input sequences into feature vectors:

- embedding_1: An embedding layer that takes in the first protein sequence (input_dim=VOCAB_SZ) and outputs a vector of shape (None, MX_SEQ_LEN, EMBED_SZ).
- embedding_2: Similar to embedding_1, but for the second protein sequence.

3) Positional Encoding Layers

Two positional encoding layers are applied to each embedding layer:

- positional_encoding_1: An instance of the Positional Encoding that takes in the embedded first protein sequence and add positional information as described in [1], using Eq. (1) and (2).
- positional_encoding_2: Similar to positional_encoding_1, but for the second protein sequence.

This positional information is calculated using following formulas:

$$PE_{(pos, 2i)} = \sin(pos / 10000^{2i / d_{model}}) \quad (1)$$

$$PE_{(pos, 2i+1)} = \cos(pos / 10000^{2i / d_{model}}) \quad (2)$$

4) Decoder Layers

There are 8 decoder layers as shown in Figure 2, each of which processes the sequence one step at a time using a same structure. Every decoding step considers both the positional encoding for both proteins and the prior output from the associated previous layer.

5) Output Layer

This is the final layer that provides us the predicted PPI score.

- A final dense layer is used to produce the model's output.
- The number of units in this layer is set to 1.
- An activation function (sigmoid) is applied to the output.

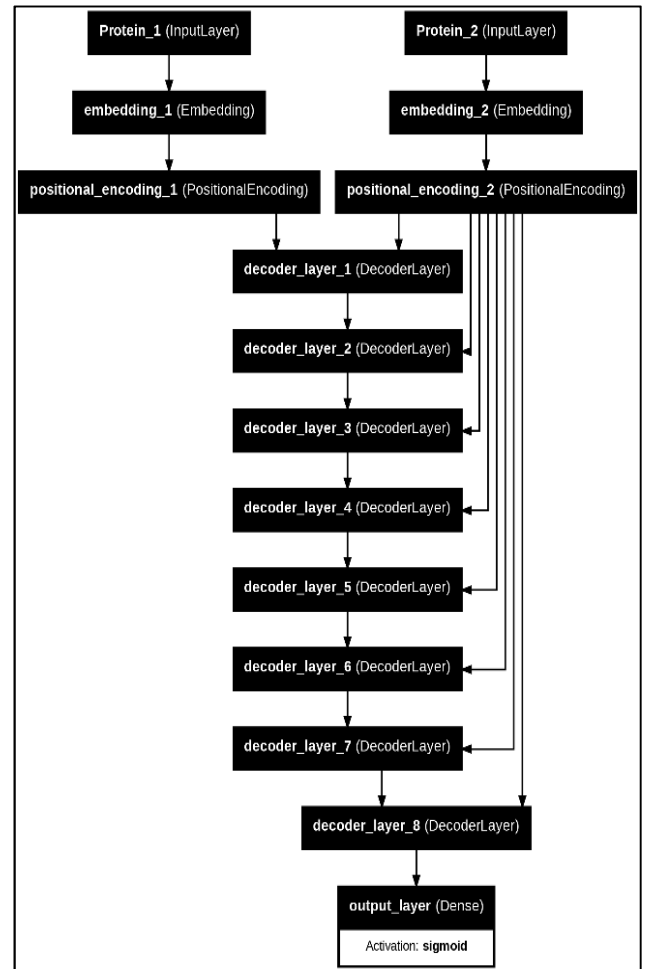


Fig. 2. Model Architecture

6) Interpreting Model Output

The DT-PPI-PM, M , takes two proteins p_1 and p_2 , to output score s which ranges between $[0,1]$ as its outcome of sigmoid activation function.

$$I(p_1, p_2) = \begin{cases} 0, & M(p_1, p_2) < 0.5 \\ 1, & M(p_1, p_2) \geq 0.5 \end{cases} \quad (3)$$

$I(p_1, p_2)$ is interaction between given protein pair p_1 and p_2 processed by model $M(p_1, p_2)$; if 0 then no interaction else if 1 then p_1 and p_2 will interact (3).

The entire model has a total of approximately 22.6 million parameters, with around 86 MB in memory usage.

III. RESULTS AND DISCUSSION

The proposed DT-PPI-PM has been evaluated using accuracy metrics in PPI to assess its performance in predicting PPIs. Table II, which compares the accuracy and other pertinent metrics across several models, presents the results. We can learn more about each model's strength by examining these indicators.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (4)$$

In terms of accuracy measures (4), the DT-PPI-PM has attained state-of-the-art performance, with high accuracy rates that surpass or compete with other models in the field. Remarkably, the suggested model attains an accuracy rate of 84.7%, demonstrating its capacity to accurately and efficiently predict PPIs. According to these findings, the DT-PPI-PM design is useful for simulating protein interactions and has important ramifications for further study and applications in fields such as biotechnology and drug discovery.

Table II. Models result metric comparison

Ref. No.	Model Names	Metric	Metric Value
[6]	TUnA	Accuracy	0.65
[7]	word2vec	AUPRC	0.717
[8]	ISPRED-SEQ	AUC	0.82
[9]	PEPPI	AUROC	0.56
[10]	HN-PPISP	ACC	0.667
Proposed Model	DT-PPI-PM	Accuracy	0.847

IV. CONCLUSION AND FUTURE WORK

The proposed DT-PPI-PM has demonstrated significant performance in predicting PPIs with an accuracy rate of 84.7%, outperforming several existing models. This achievement is a testament to the effectiveness of our novel architecture, which leverages cross-attention mechanisms and sequence-based representations to capture intricate patterns in PPI landscapes. By successfully bridging the gap between protein sequences and interaction scores, our model has vast implications for biotechnology and drug discovery applications.

The use of transformer structures and positional encoding enables the model to effectively process sequential data, allowing it to learn complex dependencies between amino acids. Furthermore, the incorporation of self-attention and cross-attention mechanisms facilitates the modelling of long-range interactions and contextual relationships within protein sequences. These features have collectively contributed to our model's superior performance in predicting PPIs.

The DT-PPI-PM architecture is set to be expanded and improved in the future. This includes expanding the species list to capture diverse protein interactions, developing methods for integrating multi-modal representations, and exploring applications in RNA-protein interaction prediction and gene regulation analysis. The goal is to create a robust framework for predicting protein interactions, applicable across biotechnological and pharmaceutical contexts, contributing significantly to understanding complex biological systems

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